Which Error Can the District Afford?

Unit 3 · Topic 3.8 · TODAY'S PROBLEM

A district checks an emergency-alert system. Let p be the proportion of alerts that arrive late. It tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, replacing the system if it rejects H_0. Both plans use $n = 200$. The Safety office wants to avoid keeping a system with a 10% late rate. The Budget office wants to avoid replacing a system with a 5% late rate. The offices have not agreed how to compare these costs.

Dev: "Use A. Smaller α lowers both error risks."

Lin: "Use B. Its higher power has no cost when safety matters."

Test plans ($n = 200$)

Plan	α	**Power at $p = 0.10$**
A	0.010	0.748
B	0.050	0.877

FIRST TAKE (5 min, on your sheet)

Which plan would you try first? Name the office you prioritized and one competing cost.

DOK 1 **(a)** Identify what the table's power at $p = 0.10$ means in this setting.

DOK 2 **(b)** Calculate the Type II error probability β for each plan. State each plan's Type I error probability.

DOK 3 **(c)** In 4–5 sentences, recommend a plan for one office using both error risks. Explain the consequence of each error and what Dev and Lin each overlook. State why the other office might choose differently.

Video + follow-along: `u6_lesson7_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.